AI-Powered Price Suggestion Mechanism

Digital Tax and Product Tracking System · Rwanda Revenue Authority

This document describes how AI can be integrated into the digital tax system to automatically generate a recommended selling price range for each product. The goal is to prevent merchants from underreporting prices for tax purposes, while preserving reasonable pricing flexibility. The AI builds a defensible range from multiple independent data sources so that no single input can be manipulated.

SYSTEM PIPELINE

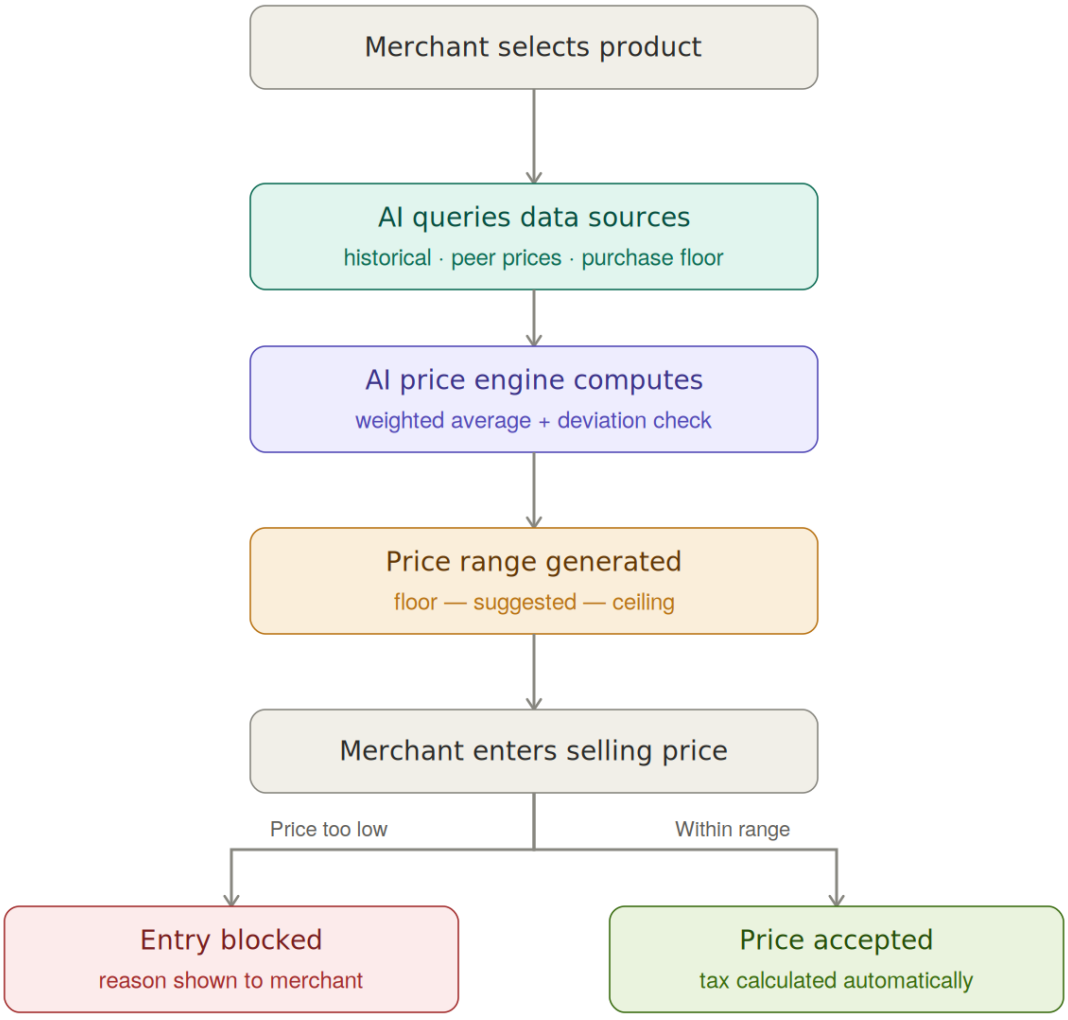


Figure 1 — AI price suggestion pipeline: from product selection to tax recording

DATA SOURCES

Source	Weight	Description
Historical prices	40%	Rolling 90-day median of declared prices for this product in this district
Peer prices	40%	The 25th–75th percentile range of what every other merchant in the sa
Purchase price floor	20%	A hard mathematical constraint: the merchant's known purchase cost (

PRICE RANGE FORMULA

The AI combines the three sources into a single suggested price and then derives a floor and ceiling:

```
suggested_price = (0.40 x historical_median) + (0.40 x peer_median) + (0.20 x floor_anchor)
```

```
floor = max(purchase_price x 1.05, suggested_price x 0.80)
```

```
ceiling = suggested_price x 1.25
```

The merchant may enter any price between **floor** and **ceiling**. A price below the floor is blocked and cannot be submitted. A price above the ceiling is allowed but flagged to RRA for optional verification, since an unusually high price does not reduce tax revenue.

MERCHANT DASHBOARD — PRICE ENTRY UI

When a merchant selects a product and prepares to enter a selling price, the dashboard displays the AI-generated range visually. The colour-coded bar shows three zones: blocked (below floor), accepted (within range), and flagged (above ceiling). Tax is recalculated live as the merchant types.

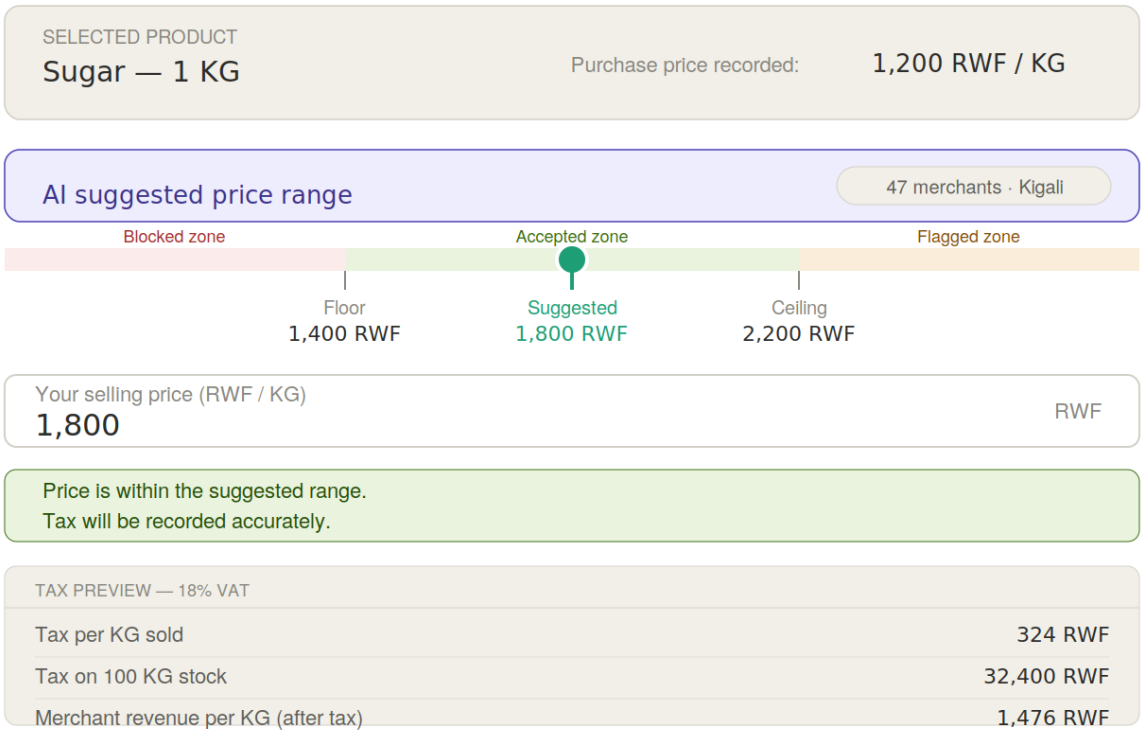


Figure 2 — Merchant price entry screen showing AI suggested range, status feedback, and live tax preview

KEY DESIGN DECISIONS

Why a floor AND a ceiling?

The floor prevents underreporting — the core problem. The ceiling prevents a different fraud where merchants declare an impossibly high price to support a later deduction claim. Both bounds are necessary for a complete defence.

Why not block everything below the suggested price?

That would be too rigid and unfair. A rural merchant may legitimately sell cheaper than a Kigali merchant due to lower overheads or local competition. The 80% floor allows a realistic 20% downward deviation before triggering a block, preserving legitimate flexibility while catching deliberate underreporting.

How are new products handled?

For products with fewer than 5 historical price records, the system falls back to a pure cost-plus estimate: floor = purchase price x 1.05, ceiling = purchase price x 2.0. The AI label changes to 'Estimated — limited data' so merchants know the range will improve as more transaction history accumulates.

What does the merchant see when blocked?

The error message must be specific and educational — not just 'price rejected'. It must say: The minimum price for Sugar in Nyarugenge is 1,400 RWF, based on your purchase cost (1,200 RWF) and market data from 47 local merchants. Please enter a price at or above 1,400 RWF. This feels fair, not punitive, and reduces merchant disputes.

How does the range stay current over time?

The AI recalculates ranges every 24 hours using fresh transaction data. If market prices rise due to a supply shortage, the suggested range rises automatically. Merchants are never locked into stale prices, and the system self-calibrates against seasonal or structural price changes without manual intervention.

IMPLEMENTATION SUMMARY

Component	Technology	Update frequency

Historical price model	Rolling 90-day median query on transaction DB	Daily
Peer price benchmark	Percentile aggregation across district merchants	Daily
Purchase floor anchor	Hard constraint from locked restock record	Per transaction
Range computation	Weighted average formula (server-side)	On product select
Merchant UI feedback	Live colour-coded bar + status message	On price keystroke
RRA visibility	Flagged entries visible on RRA central dashboard	Real-time

By combining three independent data sources and enforcing a mathematically grounded price floor, the AI price suggestion mechanism ensures that tax underreporting through price manipulation becomes structurally difficult — without being unfair to merchants who have legitimate reasons for competitive pricing.